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Article

Characterization of Per- and Polyfluoroalkyl Substances in Drinking Water Sources in the Greater Montreal Area, Quebec, Canada

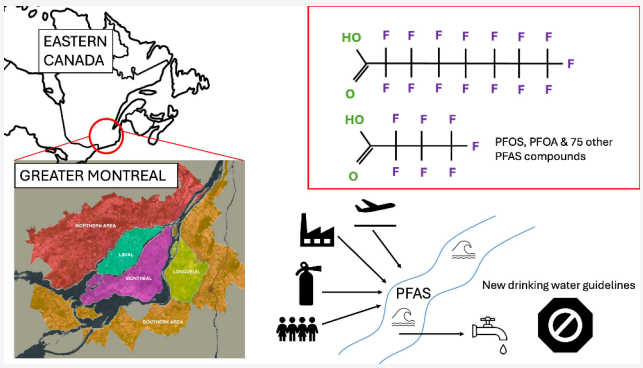
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ABSTRACT: Per- and polyfluoroalkyl substances (PFAS) are persistent organic pollutants that are subject to increasingly restrictive regulations. This study characterized the occurrence of 77 PFAS compounds in raw and treated water from 15 drinking water treatment plants (WTPs) in the Greater Montreal Area, including an urban creek receiving airport runoff. A total of 32 compounds were detected at least once, representing diverse classes and carbon chain lengths. This helped to identify trends and precursor impacts on the PFAS profiles. Perfluoroalkyl carboxylic acids (PFCA) and perfluoroalkyl sulfonic acids (PFSA) were the most frequently detected. The highest concentrations occurred in WTPs drawing from the St. Lawrence River, while the Ottawa and L'Assomption Rivers demonstrated the occurrence of localized contamination. Conventional treatment showed negligible PFAS removal. WTPs drawn from the same water source were generally correlated. Correlation analyses also demonstrated that some plants are influenced by both the Ottawa and St. Lawrence Rivers. Airport-related PFAS compounds, such as those from aqueous firefighting foam and hydraulic fluids, were detected in downstream WTPs. Seasonal trends suggest that temperature and flow variations might affect PFAS concentrations. These findings illustrate the challenges when protecting water sources against PFAS at a basin scale while offering insights into how their patterns can assist with the identification of local contamination sources.



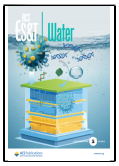
KEYWORDS: *perfluoroalkyl substances, fingerprinting, AFFFs, surface water, drinking water, urban runoff, Canada*

1. INTRODUCTION

Per- and polyfluoroalkyl substances (PFAS) are synthetic and highly fluorinated aliphatic substances, containing one or more carbon atoms, whereby hydrogen atoms on the carbon chain have been replaced by fluorine atoms.[1] Their unique characteristics have made them key parts of products such as cosmetics, nonstick kitchen utensils and tools, food packaging, waterproof clothing, and firefighting foams, among others.[1,2] PFAS compounds are stable, making their elimination from drinking water difficult due to their high persistence and solubility in water.[1,3] To reduce environmental PFAS contamination, the United Nations via The Stockholm Convention on Persistent Organic Pollutants prompted signatory states to take measures to restrict these substances.[4] The study of these contaminants has increased because of their ability to accumulate in the environment, their complex persistence and removal, and the health risks associated with exposure.[5] Cases of cancer, fertility problems, hormonal malfunctions, and developmental issues have been linked to PFAS exposure.[6] PFAS substances have been measured in the Great Lakes,[7,8] in Quebec waters,[9] including surface and drinking waters in 376 municipalities, and along the St. Lawrence estuary and Gulf.[10,11]

Knowing the scope of PFAS contamination in the Greater Montreal Area and their removal through drinking water treatment adds to the body of knowledge to assist with decision-making and regulations to limit and control their presence in the environment. The "Objective for Canadian Drinking Water Quality Per- and Polyfluoroalkyl Substances" recommends that the sum of 25 PFAS compounds be less than 30 ng/L.[12] The USEPA has restricted the concentration of PFAS in drinking water, focusing on 5 compounds (perfluorooctanesulfonate (PFOS), perfluorooctanoic acid (PFOA), perfluorohexanesulfonate (PFHxS), perfluorononanoic acid (PFNA), and GenX (hexafluoropropylene oxide dimer acid)) and a mixture of the last 3 compounds plus perfluorobutane-

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sulfonic acid (PFBS).[13] The maximum contaminant concentrations for these compounds are 4 ng/L for PFOS and PFOA and 10 ng/L for PFHxS, PFNA, and GenX.

New regulations entail economic costs due to the need to adapt or rehabilitate treatment plants to meet these new challenges. For example, the USEPA has estimated an annual cost of between US $772 million and US $1.2 billion for the elimination of PFAS in drinking water.[14] However, the American Water Works Association (AWWA) has gathered information showing that the annual cost of treatment systems to remove PFOA and PFOS from drinking water exceeds US $3.8 billion.[15,16] In April 2024, the USEPA added PFOA and PFOS to the list of hazardous substances under the Superfund program.[17] Meanwhile, 180 contaminated sites have been identified on the Superfund list across the US as being contaminated by PFAS substances including air bases and steel mill sites draining to the transboundary waters of the Great Lakes.[18]

The main objectives of this study were to (1) characterize PFAS substances in raw drinking water in the Greater Montreal Area, their temporal variations, and correlations, (2) determine the removal of PFAS substances by existing drinking water treatment infrastructure, and (3) assess the relative importance of regional and local contamination sources and the role of mixing of these large river systems on PFAS concentrations. Major rivers like the St. Lawrence, Ottawa (composed of the des Prairies and Mille-Îles), and L'Assomption Rivers were sampled for representativeness and geographical coverage. Health Canada recommends that municipalities and provinces characterize their sources to apply measures, modify, or improve drinking water treatment. The recommendations are aimed at areas likely to be contaminated with or contain considerable concentrations of these compounds.[12] Montreal and its metropolitan region represent nearly 4 million inhabitants with many industrial activities as potential sources of PFAS contamination.[19,20] The results were compared with the guidelines and objectives for PFAS in drinking water in both Canada and the US. This study analyzed the occurrence of PFAS, their characteristics, potential origins, and transformation in the environment. An urban creek was sampled to evaluate the potential impact of airport runoff on a downstream drinking water treatment plant and compare regional PFAS contamination with local sources.

2. MATERIALS AND METHODS

2.1. Sample Collection. Samples of raw water (n = 32) and treated water (n = 28) were collected on two sampling campaigns in February and May 2023. The same treatment plants were sampled each time. Samples were taken in pairs (raw and treated water), except for one site, where only treated water was sampled because a representative raw water sample was not possible. A water treatment plant on the St. Lawrence River (A5) was also monitored weekly from August to November 2023 to assess higher frequency temporal variations. The urban creek was sampled in May and August 2023. Except for this last sampling site, all water samples were collected from drinking water treatment plants (WTPs) (Figure S1 in the Supporting Information (SI)). Raw water samples were taken as close as possible to the intake point, while treated water samples were collected from the final distribution line outlet after all treatment steps (more information on water sample collection is found in Text S2 in SI). This paired sampling approach ensured that both samples reflected water processed through the same treatment cycle, enabling a reasonable estimation of PFAS removal efficiency across each plant. Censored values were treated as zeros.

During sampling, various physicochemical characteristics were measured using standard methods,[21] including water intake type (main or secondary), pH, temperature, turbidity, UV absorbance (UVA_{254}), and total organic carbon (TOC). All parameters except TOC (analyzed at the Polytechnique Montréal) were analyzed by the WTPs. Additional long-term data collected through routine monitoring, such as alkalinity, were provided by the participating utilities.

2.2. Chemical Analysis. Each water sample was filtered (1 L) using Advantec glass fiber filters (GF-75, 0.3 μm, 47 mm diameter) from Sterlitech Corporation (Kent, WA, USA). The filtered material was retained for analysis of PFAS sorbed on the particulate phase. For each sample, two extractions were performed: a solid-phase extraction or SPE (Text S2 in SI) for the dissolved phase and an ultrasonic extraction for the particulate phase. The first extraction was performed using an automated off-line solid-phase extraction (Thermo Dionex Autotrace SPE 280) and weak anion exchange cartridges (Strata X-AW). In the case of suspended particles, ultrasound-assisted solvent extraction was used. Concentrated extracts were diluted in HPLC water on a manifold before being subjected to SPE (Strata X-AW cartridges).[11] All analyzed substances were subjected to ultrahigh-performance liquid chromatography (UHPLC), connected to a Q-Executive Orbitrap mass spectrometer (Thermo Scientific, Waltham, MA, USA), which was operated in full scan mode.[11]

2.3. Quality Control. The SPE process was carried out on 6 bottles a day, which included 2 blanks to prevent cross-contamination during sampling or in the laboratory (Text S2 in SI). The handling of blanks was identical to the preparation of field samples.

2.4. Statistics. Statistical analyses were conducted using R statistical software.[22] A Wilcoxon nonparametric test was used to compare raw and treated water. Principal coordinate analysis (PCoA) comprising a multivariate homogeneity analysis of group variances (betadisper),[23] ANOVA tests, and a permutational multivariate analysis of variance were used to compare among groups. Principal component analysis (PCA) was used to find correlations among different variables in all of the water samples. A Bayesian and multiplicative substitution method was applied for count zeros, while a Hellinger transformation was performed to normalize the data prior to PCA. The threshold of statistical significance (p-value) was set at 0.05. Package ggplot2 in R was used for data visualization.[24]

2.5. Sampling Sites. Greater Montreal's hydrographic network covers 15% of the territory and serves as its primary source of drinking water.[25,26] The sampled network consists of the St. Lawrence River (8 sampling sites, 1 sampled for treated water only, category A), the Ottawa River (des Prairies and Mille-Îles, 4 sites, category B) and L'Assomption River (3 sites, category C). Sampling sites are drinking WTPs, located in 6 municipalities in the area. The WTPs are labeled in the direction of water flow; for example, C1 is upstream of C2, and C2 is upstream of C3. An urban creek (U1), which receives airport runoff, was also sampled during the spring season.

3. RESULTS AND DISCUSSION

3.1. PFAS Substances, Temporal Variations, and Correlations in Raw Drinking Water. *3.1.1. Occurrence*

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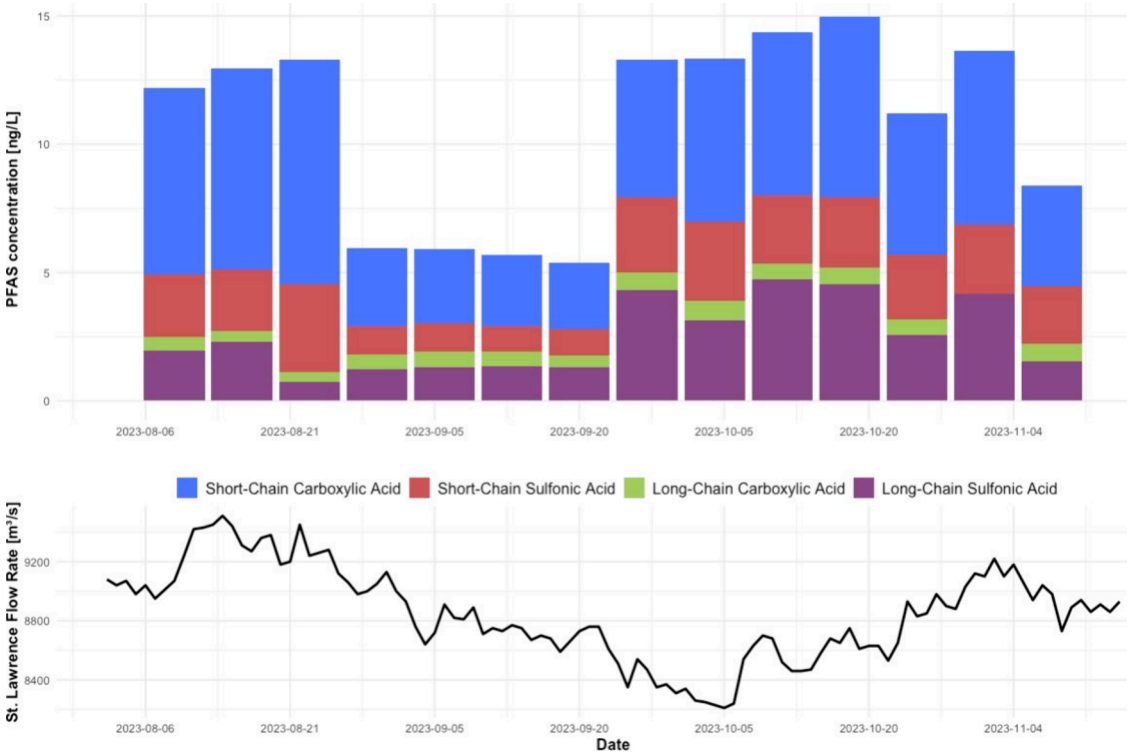


Figure 1. Temporal variations in PFAS concentrations by main acid type (carboxylic and sulfonic), chain length (short and long), and flow rates for the St. Lawrence River. Flow rate data were collected from Environment and Climate Change Canada.[34]

in Raw Waters. PFAS can be categorized both by class and by carbon chain length. The literature differs on how to classify short- from long-chain PFAS structures. For instance, Wang et al.[2] state that perfluoroalkyl carboxylic acids (PFCA) with more than 7 perfluorinated carbons and perfluoroalkyl sulfonic acids (PFSA) with more than 6 perfluorinated carbons are long-chain PFAS. In this study, we have identified all PFAS compounds with 6 or more perfluorinated carbons as long-chain PFAS, and the remainder are categorized as short-chain.

In raw water samples, of the 77 targeted compounds (Text S1 and Table S1 in SI), 24 were above detection limits in the winter (Table S2 in SI). In the spring, 27 compounds were identified in raw water (Table S2 in SI). PFAS were present in suspended solids, although the concentrations were significantly lower compared with the dissolved phase. In the winter, long-chain PFAS in suspended solids accounted for 62.5% of the total concentration and decreased to 32.6% in the spring. This decline may be attributed to the increased dilution of contaminants and effluents, leading to an increase in the suspended solids. Previous studies[27] have demonstrated the high percentage (approximately 80%) of long-chain PFAS in suspended solids.

Features of the study area such as upstream industries, hydrographic characteristics, influence of flagged US Superfund sites, wastewater treatment plants (WWTPs), and other potential sources of PFAS contamination of these surface waters are provided in Texts S3 and S4, Table S3, and Figure S2 and provide an explanation for the contamination demonstrated by other studies.[28]

All measured total concentrations were below the Health Canada guideline value of 30 ng/L. GenX was not detected at any site, regardless of the season. In 2024, the USEPA[13] also introduced a Hazard Index approach for the combined toxicity of PFHxS, PFNA, GenX, and PFBS in mixtures. As several of these compounds were codetected in samples across sites, the cumulative presence of PFAS, even at low individual concentrations, may still warrant attention in future risk assessments or regulations. PFOA is one of the most abundant compounds in winter and spring, where its maximum concentration was 1.64 ng/L. Its presence throughout the seasons was expected, as PFOA and PFOS are considered legacy PFAS, persistent in the environment.[2] As with PFOA, the maximum PFOS concentration was also detected in spring with a value of 1.28 ng/L. PFNA, PFHxS, and PFBS have maximum concentrations of 0.31, 0.92, and 1.03 ng/L respectively. Similar patterns of occurrence have been reported in various US states, including Pennsylvania,[29] Massachusetts,[30] Nevada,[31] and Florida.[32]

3.1.2. Temporal Variations. Water quality varies seasonally, as do PFAS concentrations.[33] Spring is characterized in this watershed by higher flows,[34] which can dilute contaminants or facilitate the conveyance of local contaminant sources to drinking water intakes. Urban, industrial, and agricultural activities have a direct impact on water quality as sources of chemicals and nutrients affecting ecosystems.[35] Previous studies have analyzed temporal variations of PFAS substances with sparse results. For example, a decrease of PFAS in sediments from autumn to summer was observed, while PFAS in surface water remained constant throughout the year.[36] Brown et al.[37] examined temporal variability over 5 weeks, noting a decrease of most PFAS concentrations following heavy rainfall, indicating dilution from stormwater. However, an increase was observed for a subsequent event.

To explore temporal variations in the St. Lawrence River, the A5 plant was monitored weekly from August to November of 2023 with flow rates greater than 8,200 m^3/s. The large flow rates of the St. Lawrence River mean that local stormwater inputs have a lesser impact on flow rates and concentrations of

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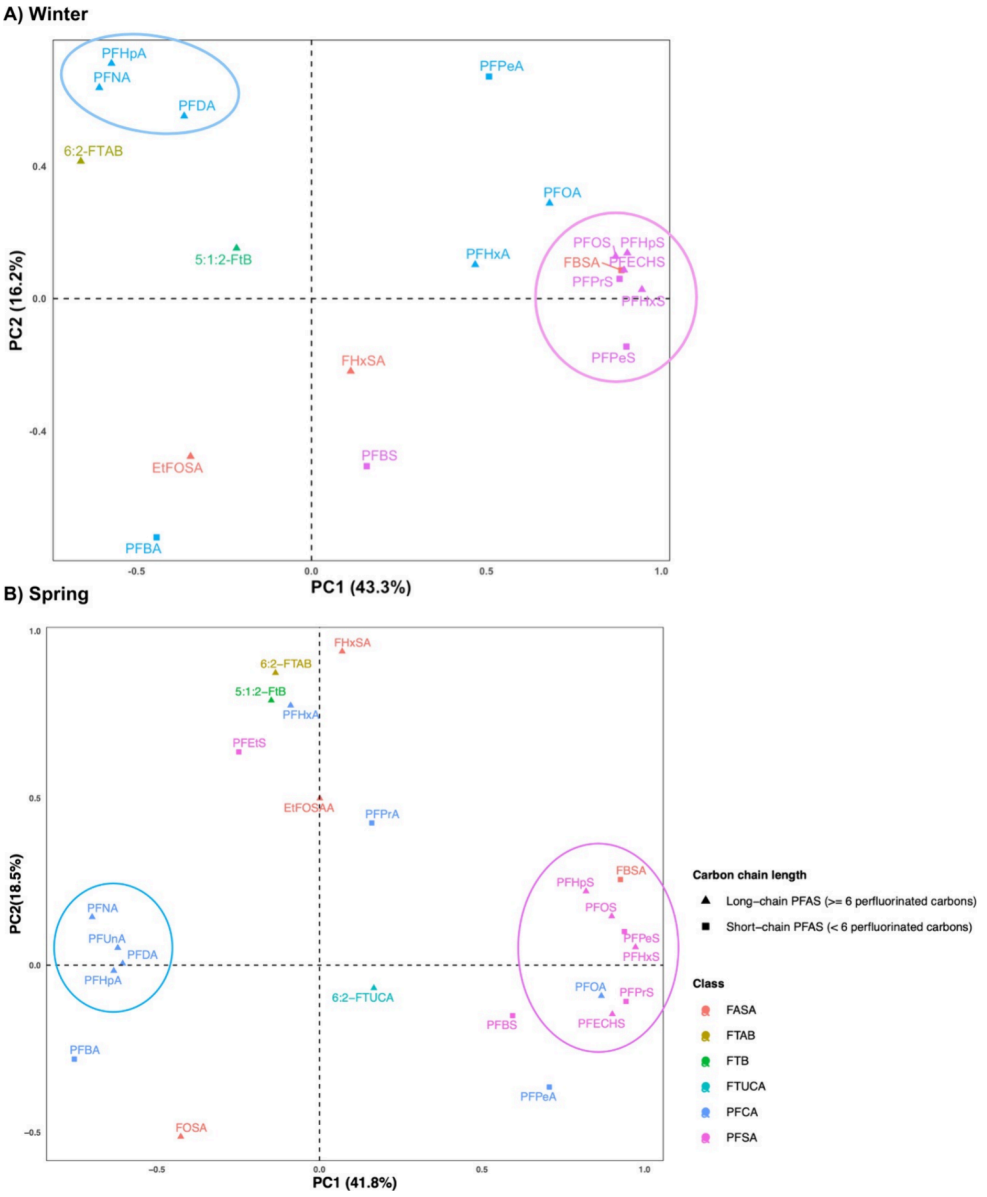


Figure 2. PCA of PFAS compounds in raw water samples collected during winter (A) and spring (B), showing detected compounds, their chain lengths, and corresponding classes: FASA (perfluorooctanesulfonamides), FTAB (fluorotelomer alkylbetaine sulfonamide and perfluoroalcoholic betaines), FTB (fluorotelomer betaine), FTUCA (fluorotelomer unsaturated carboxylic acids), PFCA (perfluoroalkyl carboxylic acids), and PFSA (perfluoroalkyl sulfonic acids).

highly persistent compounds. Samples were collected following treatment by an Actiflo[38] and sand-anthracite filtration. However, as illustrated in Figure 3, conventional treatment is not effective for PFAS removal, and variations within these samples were considered as representative of variations of source water quality.

In Figure 1, compounds are separated according to their main acids (carboxylic and sulfonic) and chain length. In general, concentrations of short-chain PFAS are higher than their long-chain counterparts.[39] Total concentrations of PFAS remained within an approximate range of 5 to 15 ng/L with highest total concentrations occurring during and immediately following the lowest flow period of the St. Lawrence,[34] a potential indication of the relative importance of dilution processes at large regional scales for these highly persistent compounds. The total and relative concentrations of the short-chain carboxylic group were highest during the summer. In contrast, the total and relative concentrations of long-chain sulfonic acids were highest during and immediately following the period of lowest flow (Figure 1). Gago-Ferrero et al.[40] observed an increase in the concentration of micropollutants, including PFAS, in the summer months for WWTPs. In contrast, Nguyen et al.[39] linked an increase of PFAS concentrations with periods of high flow such as floods or snowmelt. Temporal trends, especially for large river systems, reflect a wide variety of regional and local scale sources and processes that include dilution of distant industrial sources, atmospheric wet deposition, local contributions from contaminated sites, pesticides in agricultural runoff and drainage, WWTPs, and combined sewer overflows (CSOs). Above average precipitation occurred in Montreal in July 2023 as compared to previous years[41] and could have influenced August concentrations. Upstream pesticide application,[42] industrial discharges, and atmospheric deposition can vary

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seasonally, with delayed impacts on surface water concentrations due to transport and retention mechanisms.[43,44] With longer and more extensive time series, the changes in the relative proportions of short-chain carboxylic acids and long-chain sulfonic acids could be explored further as indicators of local and regional transport processes, including the seasonal influence of wastewater inputs and agricultural drainage. Faunce et al.[45] presented a method to calculate the varying proportions of streamflow attributed to wastewater in the Potomac River, offering a methodology to quantify PFAS loads at a regional scale. While high flow events can dilute persistent contaminant concentrations, low flow periods can increase the proportion of streamflow originating from wastewater discharges and agricultural runoff and field drainage, thereby increasing concentrations at specific times of the year.

3.1.3. Correlation among PFAS Compounds. PFAS compound diversity was mainly focused on PFCA (perfluoroalkyl carboxylic acids), PFSA (perfluoroalkyl sulfonic acids), FASA (perfluorooctanesulfonamides), FTAB (fluorotelomer alkylbetaine sulfonamide and perfluoroalcoholic betaines), FTB (fluorotelomer betaine), and FTUCA (fluorotelomer unsaturated carboxylic acids), which were detected in all water samples. As observed in Figure 2A,B, PFSA are mostly grouped together, except for PFBS, the short-chain version of PFOS. There is a strong correlation among PFOA, PFOS, and PFHxS (former PFAS considered persistent organic pollutants by the Stockholm Convention). Correlations of PFOS with other PFSA compounds such as PFHxS and PFHpS has been observed in human biomonitoring.[46] Thus, a correlation between PFOS and PFOA was expected, as shown in Figure 2A,B, since they share sources and environmental persistence,[2] have similar environmental behavior,[47] or co-occur with other pollutants.[10,48] However, Nøst et al.[46] also showed a correlation between PFOS and PFCA compounds such as PFNA and perfluorodecanoic acid (PFDA), a trend we did not observe. PFNA and PFDA were strongly correlated, which was also observed by Nøst et al.[46] No correlations associated with the chain length were observed. The PFAS with the longest chains above detection limits were 5:1:2 fluorotelomer betaine (5:1:2-FTB) and 6:2 fluorotelomer sulfonamide alkylbetaine (6:2-FTAB), 12 and 14, respectively.

A negative correlation between the main PFSA compounds and 6:2-FTAB was observed in the winter and spring (Figure 2). The compound 6:2-FTAB was created as an alternative to sulfonated PFAS, such as PFOS,[49] and therefore one would expect a positive correlation between them. However, the negative correlation in this study could be related to the sources of the compound and its environmental fate. 6:2-FTAB is subject to photodegradation in waters exposed to sunlight;[50] thus, its presence could indicate a local source of contamination. Aqueous film-forming foams (AFFFs) are a source of 6:2-FTAB, which can end up in WWTP effluent[51] upstream of WTPs. WWTPs have a low capacity to remove 6:2-FTAB.[52] Perfluorobutane sulfonamide (FBSA) has been detected in tap water along with perfluorohexane sulfonamide (FHxSA)[8] in Canadian surface waters, and in trout from the Great Lakes.[54] FBSA could be a PFAS decomposition compound present in some AFFF foams and surfactants.[53]

Although it was detected in low concentrations, the highest concentration of Sulfluramid (EtFOSA) was observed in the winter at an Ottawa River WTP with urban and agricultural land use immediately upstream. Its occurrence might be linked to agricultural activity, as it is an active ingredient in commercial pesticides and can break down into PFOS,[55] posing regulatory concerns as pesticide-derived PFAS represent a less monitored input pathway to surface water.[56] However, considering the prominent urban and residential land use in this region with wastewater effluents potentially contaminated by residential-use pesticides,[56] a mixture of contaminant sources is likely and warrants further investigation. Perfluoroethylcyclohexanesulfonic acid (PFECHS) concentrations have been measured in the Great Lakes, although its origin in this region is unclear, as it may be caused by a discharge event or by emissions over the years.[7] FBSA, FHxSA, and PFECHS have been detected in the St. Lawrence River.[57]

De Silva et al.[7] found no statistical correlations between PFOS/PFOA and PFECHS in surface waters. However, we observed (Figure 2) a strong correlation between PFECHS and most legacy PFAS. This is consistent with the origin of PFECHS, which appears to be linked to other legacy PFAS.[58] PFECHS contamination might be linked to airports, as the compound is typically detected in water bodies close to airports even in remote and isolated areas around the world.[58]

In the spring (Figure 2B), there was a strong correlation among PFSA compounds, as they are grouped together regardless of their chain length. However, the PFCA compounds were weakly distributed. Long-chain PFCA (PFDA, perfluoroundecanoic acid (PFUnA), PFNA, and perfluoroheptanoic acid (PFHpA)) are grouped together, while short-chain PFCA are distributed in the plot (perfluoropropionic acid (PFPrA), perfluoropentanoic acid (PFPeA), and perfluorobutanoic acid (PFBA)). PFPeA, a short-chain PFAS, was strongly correlated with the other PFSA. Perfluorohexanoic acid (PFHxA) is strongly correlated with compounds considered to be precursors and can be formed through the degradation of longer-chain PFAS.[59] While absent in winter (Figure 2A), the compound *N*-ethyl perfluorooctane sulfonamido acetic acid (EtFOSAA) was detected in spring. It has also been detected previously in surface water, and its appearance could be linked to the transformation of perfluorooctane sulfonamide precursors.[60] Wastewater studies have shown that the compound EtFOSAA is a degradation product of the compound perfluorooctane sulfonamidoethanol (EtFOSE), and at the same time, it can degrade into EtFOSA. The latter will eventually decompose once again to become perfluorooctane sulfonamide (FOSA), which will transform again until it becomes PFOS.[61]

Although the concentration is relatively low (0.04 ng/L), the compound perfluoroethanesulfonic acid (PFEtS) was detected in Ottawa and L'Assomption River sampling sites (Figure 2B). The Ottawa and L'Assomption Rivers have mixed land use (forested/agricultural/urban), with 73%, 3%, and 1%, respectively, for the Ottawa Basin and 68%, 17.7%, and 5.7% for the L'Assomption Basin.[62,63] Land use can affect PFAS concentrations.[29] PFEtS is linked to AFFF foam production. One of the characteristics of this compound is that it appears to be as persistent as its long-chain counterparts, since it has been associated with the contamination of military sites used for training purposes in the United States.[64]

3.2. Removal of PFAS Compounds by Drinking Water Treatment. PFAS substances are difficult to remove from drinking water. According to Zhao et al.,[65] their low potential for adsorption prevents PFAS from attaching to suspended solids; thus, settling is not an effective removal process during conventional treatment. Their high solubility, reduced

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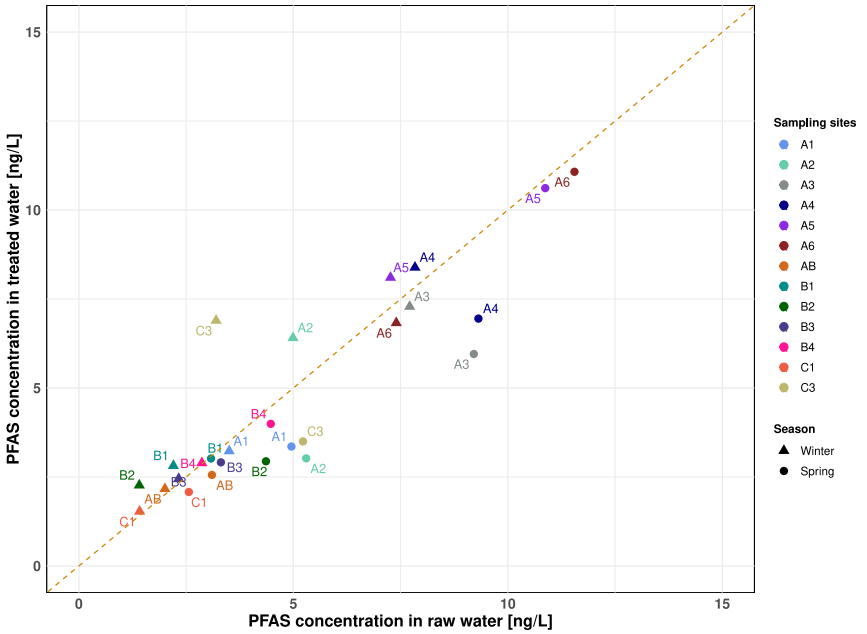


Figure 3. Low PFAS removal by treatment plants in the winter and spring. Category A WTPs have their water intakes in the St. Lawrence River, while those of B and C WTPs are in the Ottawa and L'Assomption Rivers, respectively. The concentration for plant A7, which has only treated water, was 6.7 and 4.8 ng/L in the winter and spring, respectively. For site C2, the winter raw water concentration was 3.6 ng/L, while the spring treated water concentration was 4.7 ng/L.

volatility, and simultaneous hydrophobicity and hydrophilicity facilitate their occurrence in aqueous environments.[66]

While PFAS profiles in raw and treated water were generally similar, some differences were observed. In treated water during the winter and spring, 28 compounds were above detection limits. The compounds 8:2 fluorotelomer sulfonic acid (8:2-FTS) and 6:2 fluorotelomer carboxylic acid (6:2-FTCA) were detected only in raw water. In contrast, several compounds were detected exclusively in treated water, including PFEtS, methylperfluorooctanesulfonamide (MeFOSA), perfluorodecanesulfonamide (FDSA), perfluoro(2-ethoxyethane)sulfonic acid (PFEESA), and 6:2 unsaturated carboxylic acid fluorotelomer (6:2-FTUCA), potentially caused by transformation during treatment.[65,66] As for the treated water in spring, additional detections included perfluorotridecanoic acid (PFTrDA), 8:2-FTS, 4,8-dioxa-3*H*-perfluorononanoic acid (ADONA), 5:3 fluorotelomer betaine (5:3-FTB), and perfluoro-4-methoxybutanoic acid (PFMBA), while 6:2-FTCA, 6:2-FTUCA, MeFBSA, and perfluoro-3,6-dioxaheptanoic acid (3,6-OPFHpA) were not above detection limits. Despite these variations, all WTPs would comply with the current regulatory limits of the USEPA. However, as seen in Figure 3, the removal capacity of PFAS in treatment plants remains low, which was expected, given the existing infrastructure and treatment technologies (Table S4 in SI). Differences in PFAS removal across treatment plants may be partially explained by the treatment technologies employed (Table S4). To compare removal among plants, a Wilcoxon test was conducted, resulting in no statistical difference in the PFAS removal across WTPs. However, when comparing removal by PFAS compound, WTPs that include only ozonation and filtration (A2, AB, and B2) generally show limited PFAS removal, which is consistent with previous studies indicating that ozonation and UV processes are ineffective for most PFAS compounds.[66,67] In contrast, plants using activated carbon (such as A6, B4, and C3) tend to show greater, though still almost negligible, reduction in long-chain PFAS. However, the overall magnitude of removal remains modest, and variable outcomes between plants with similar technologies suggest that other factors, such as media age, contact time, or operational conditions,[66] may also influence performance.

Removal capacity ranged from 0.3 ng/L (8% removal rate) to around 3 ng/L (36% removal rate), consistent with the results from previous studies. Takagi et al.[68] reported that PFOS and PFOA were detected in both raw and treated tap water, with similar concentrations before and after treatment, indicating minimal removal during conventional treatment processes. Quiñones and Snyder[69] also found persistent levels of PFOS and PFOA in a US WTP, highlighting the persistence of these compounds in conventional drinking water treatment. Furthermore, Takagi et al.[70] demonstrated that ozonation and sand filtration had negligible effects on PFOS and PFOA removal, while activated carbon was only effective if recently installed, losing effectiveness after a year of use. These findings reinforce that physicochemical characteristics of PFAS compounds result in their near-zero elimination from drinking water (Text S3 in SI). Consequently, the population may be exposed to PFAS concentrations like those found in raw water given that conventional treatment is inefficient for their removal.[68,71] Other factors influence the elimination of PFAS compounds, such as the length of the carbon chain functional group[72] and the type or possibility of biodegradation into other, shorter compounds.[73,74]

Winter and spring have roughly the same PFAS removal capacity. The low removal capacity of conventional drinking water treatment has compelled researchers to examine the effectiveness of point-of-use and point-of-entry residential drinking water filters to assess their PFAS removal efficiency, with positive results.[75,76] These household filters could serve as a temporary barrier to reduce exposure to these compounds until other long-term sustainable solutions are found.

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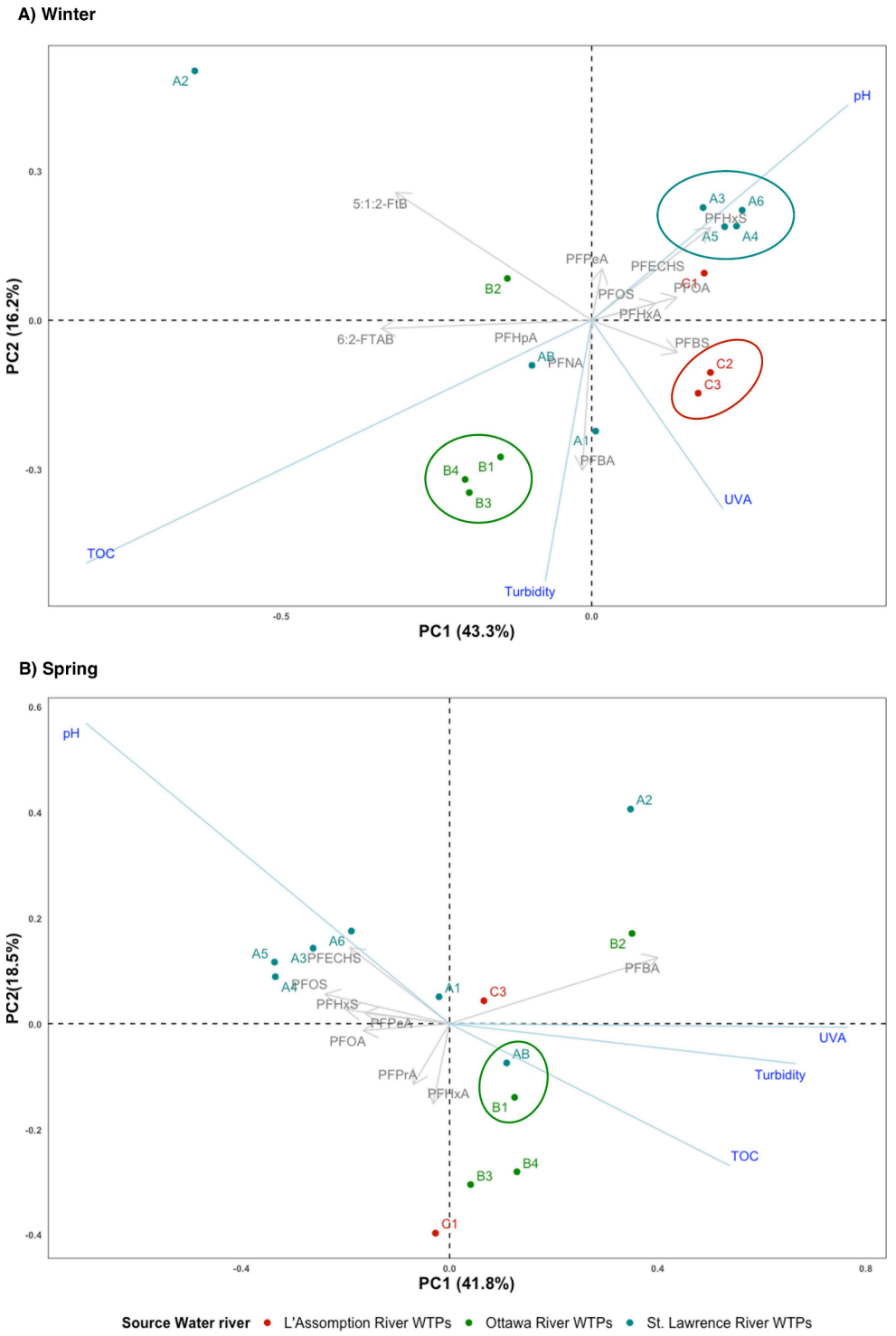


Figure 4. PCA of physicochemical parameters (pH, UVA, turbidity, and TOC) and PFAS compounds in raw water samples from drinking water treatment plants (WTPs) in three river basins (L'Assomption, Ottawa, and St. Lawrence Rivers) during winter (A) and spring (B).

3.3. Regional and Local Contamination Sources and the Role of Large River System Mixing. Given their high solubility and persistence, PFAS compounds are potential markers of water origins in diverse mixed systems. Changes in water quality resulting from changes in mixing patterns in large river systems can affect drinking water treatment. The Greater Montreal Area provides a useful test site to assess the mixing of large river systems using PFAS substances as markers because some drinking water intakes variably receive varying mixtures of water from the St. Lawrence and Ottawa Rivers.

Statistical tests were performed to find relationships among sampling sites, PFAS compounds, and their respective classes. PFAS classes were divided into perfluoroalkyl sulfonamides (FASA), alkylbetaine sulfonamide fluorotelomer (FTAB), fluorotelomer betaine (FTB), fluorotelomer unsaturated carboxylic acids (FTUCA), perfluorocarboxylic acids (PFCA), and perfluorosulfonic acids (PFSA). The PCA figures

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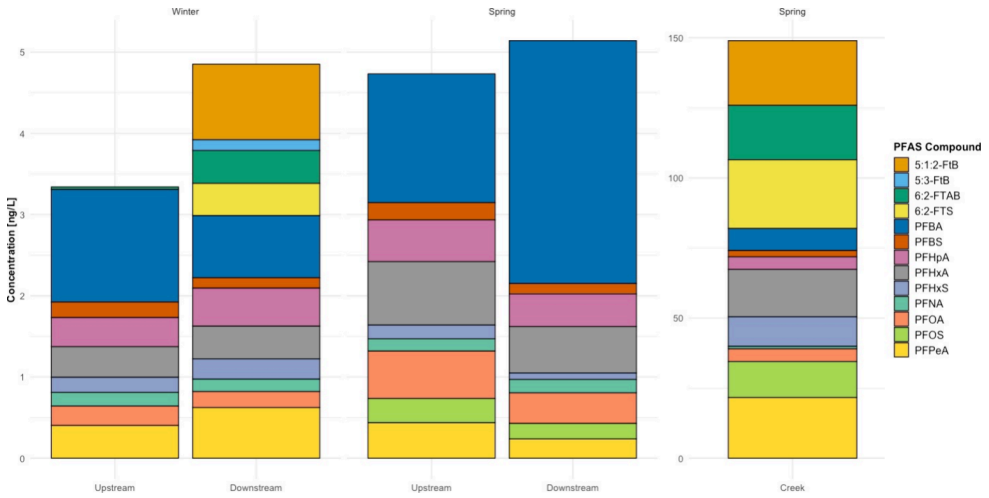


Figure 5. Impact of airport runoff on downstream WTPs. The downstream values represent the average raw water (n = 2) of the two targeted WTPs (A2 and A3). Concentrations are given in ng/L. Urban creek (U1) results were available only for spring.

illustrate the profiles of the observed PFAS compounds. Category A plants, which have water intakes in the St. Lawrence River (A3, A4, A5, and A6), were almost always grouped together in both seasons. In winter (Figure 4A), the A1 plant is more likely to be clustered with B-type plants. To understand why, we analyzed the influence of the Ottawa River and WWTPs on the water intake of plant A1.[77] Flow data from three hydrometric stations[34] were used to calculate the flow ratio between the Ottawa River and the St. Lawrence River ($Q_{Ottawa}/Q_{St\text{-}Lawrence}$) for each season of 2022. Based on flow estimates at the Lasalle monitoring point and known seasonal hydrodynamic conditions, the mixing zone between the St. Lawrence and Ottawa Rivers shifts southeast in winter, allowing water from the Ottawa River to influence the A1 intake.[78] This is supported by the clustering pattern in Figure 4A, where A1 is grouped with B-type plants drawn from the Ottawa River. Generally, there are no wastewater sources contributing simultaneously to A- and B-type plants except when the A-type plants are under the influence of the Ottawa River. This could explain why plant A1 is closer to type B plants. As seen in Figure 4A,B, the remaining plants are usually clustered with their corresponding category representing the river source.

Plant AB is grouped with sites corresponding to the Ottawa River (Figure 4B). This correlation was expected, as this plant receives a mixture of water from the St. Lawrence River and from the Lake of Two Mountains draining the Ottawa River. Furthermore, when analyzing alkalinity data between sites AB and B1, similar values are observed throughout winter and spring. Regarding plants C1 and C3, we note that they are not grouped together (Figure 4). Plant C1 is the furthest upstream from major population centers, with a smaller population in an area of agricultural activity. C3 had higher concentrations than C1, but the same compounds were detected, demonstrating the addition of local contaminant sources as the river flows.

In both seasons (Figure 4), water treatment plants A2 and B2 are not grouped with other plants on the same rivers. A2 was specifically analyzed due to its proximity to an urban creek (U1) that receives runoff from an airport. Given that A2 and B2 are grouped together in the same quadrant in both spring and winter, a potential explanation was sought for similarities between the source waters of the two WTPs. Both WTPs are immediately downstream of urban creeks draining the same airport but are in different flow directions. A hydraulic connection between creek U1 and the A2 WTP is possible based on hydrodynamic modeling.[77] Hydrodynamic modeling is not available for the creek flowing north upstream of B2. However, other studies have shown that the plumes of urban effluents are able to reach these drinking water intakes depending on hydraulic conditions.[79] Urban runoff can have an impact on PFAS pollution of water sources, even more so than WWTPs,[80] as airport runoff also impacts surface waters.[81,82] Flagged US Superfund PFAS sites[18] affecting water quality tend to be industries using PFAS in their production chain, airports, or military sites. Thus, the occurrence of specific PFAS compounds at these two water treatment plants might be associated with upstream land use, as observed elsewhere.[83] The contribution of industrial sites as PFAS sources in the Great Lakes−St. Lawrence River system is widespread and ultimately contributes to oceanic concentrations, where these persistent compounds are remobilized into the atmosphere through sea spray aerosols, perpetuating their global cycle.[84,85]

Despite the high density of CSOs and/or wastewater discharges in the river sections of the Ottawa River, PFAS concentrations in these tributaries remain comparatively low. This river section includes up to 190 CSOs outfalls and multiple WWTPs upstream of drinking water intakes.[86] A study of the impact of CSOs on raw water concentrations demonstrated that these events significantly increased *E. coli* concentrations, confirming a strong wastewater influence.[87] However, even with disproportionally high wastewater and CSO contributions in the Ottawa River as compared to the St. Lawrence River, PFAS concentrations in the Ottawa River remain lower than those observed in the St. Lawrence River despite its large flow rates. This suggests the presence of upstream contaminant sources in the Great Lakes−St. Lawrence River system might contribute to PFAS levels observed in the St. Lawrence in the Greater Montreal Area, rather than local tributaries, CSOs, or wastewater. PFAS contamination in the Great Lakes has been previously documented.[88,89] However, in some instances, local sources must also be considered, as discussed further below. Additional monitoring upstream of the Greater Montreal Area would be required to distinguish long-range transport from local contributions.

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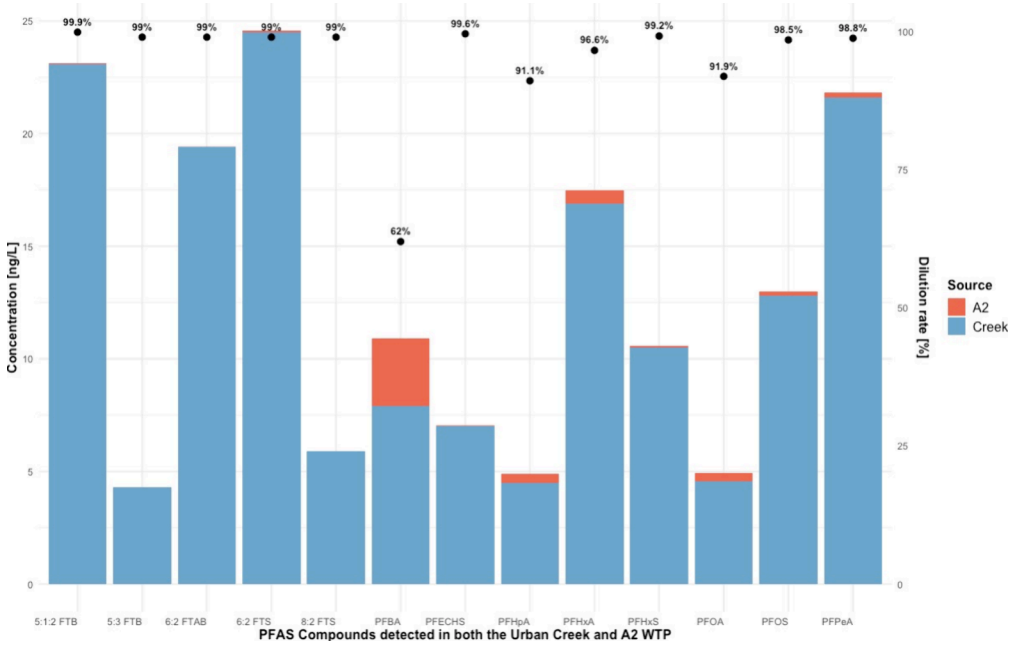


Figure 6. Dilution rate for PFAS compounds detected in the urban creek (U1) and A2 (during spring). The percentage shown is the difference between the creek and the downstream WTP.

A correlation was observed among WTPs on the St. Lawrence and the occurrence of compounds such as PFOS and PFOA. The co-occurrence of PFOS and PFOA was also observed in previous studies.[27] Figure 4A illustrates the relationship of precursors such as 5:1:2-FTB measured in winter and the A2 and B2 plants, which may indicate the influence of local contamination from urban creeks, which are impacted by urban runoff and could suggest a common source. Urban runoff is a significant pathway for precursors to enter aquatic systems,[90] which could explain the correlation. In the spring (Figure 4B), the trend for PFCA and PFSA is repeated, with the particularity that the occurrence of the compound PFBA has a strong presence at plant B2. The presence of PFBA might be explained by the intrinsic properties of water in the spring season (i.e., low temperatures, high flows) and the low degradability and strong migratory capacity of PFBA.[91]

Tributaries to the Great Lakes−St. Lawrence River system have a great impact on total PFAS concentrations.[7,8] The contrast with the Ottawa and L'Assomption Rivers whose basins are found exclusively in Canada is notable. These basins provide a portrait of contamination by local PFAS sources, which is consistent with research by Munoz et al.,[10] where PFAS concentrations were also found in water supplies in rural Quebec.

The impact of the creek (U1) on two drinking water treatment plants (A2 and A3) was explored to understand why several compounds (like 5:1:2-FTB or 6:2-FTAB) were detected at these sites but not at the upstream plant (A1), as illustrated in Figure 5. All samples shown were collected during the same sampling event to allow for direct comparison. Although PFAS concentrations in U1 were higher than for downstream locations, the PFAS profiles differ, suggesting dilution, transformation, or partitioning before reaching the downstream plants. Urban creeks and storm sewers with illicit cross-connections receiving residential wastewater in this sector have been studied previously[92] and are an additional potential source of PFAS contamination. However, U1 was identified as particularly vulnerable to PFAS contamination discharging downstream of A1 and upstream of the A2 and A3 sampling locations. Urban creek U1 is ungauged.

Concentration differences between WTPs up and downstream of the urban creek can be useful for estimating the dilution of contamination sources such as the creek flowing toward a water intake, as shown in Figure 6. The dilution of PFAS substances transported from their sources has been described in other studies.[81,93] Given the persistence of PFAS compounds, they are useful for estimating the dilution potential in the event of an accidental event or spill, such as the case in Etobicoke Creek.[82] It also provides information to interpret the temporal and spatial variability of PFAS compounds.[94]

As expected, concentrations at the downstream drinking water treatment plant (5.3 ng/L in spring) were lower than concentrations in the creek, where concentrations are on the order of 150 ng/L (Figure 5). Distance from the source of contamination (airport) influences the concentration in the plume.[95] The distance from the creek sampling point (U1) to the water intake is approximately 5 km. Figure 6 illustrates the high dilution ratio by comparing the concentrations in the creek and those at the A2 plant intake. The low concentrations of 6:2-FTAB at plant A2 may indicate its phototransformation. The nondetection of 5:3-FTB and the low detection of 5:1:2-FTB in spring at plant A2 might be explained by the ability of long-chain PFAS to partly adsorb to soils and sediments. A study of the riverbed characteristics could provide more information on PFAS present in sediments. It is important to note that apart from a few compounds related to airport activity, the other WTPs whose source water is the St. Lawrence River (category A) have PFAS concentrations that are not statistically different from A2. The low dilution rate of PFBA compared to other compounds (Figure 6) could be related to the presence and degradation of precursor compounds[96] that can increase its concentrations and affect the dilution rate.

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Although no PFAS concentrations were measured in another urban stream (U2) located upstream of plant B2 during the period of study, it also drains a portion of the airport and an industrial area, like urban creek U1. Given that A2 and B2 were grouped together in Figure 4, it is worth exploring PFAS concentrations in U2 and other discharge points. Urban runoff is a source of PFAS, including precursors or short-chain compounds.[90] The colocation of B2 with sites receiving such runoff and the similarity in compound patterns between B2 and A2 in the PCA support both streams as contributors to PFAS profiles to their respective receiving waters, but this requires further study. When comparing the compounds present in B2 with those in the upstream treatment plant (B1), we see that 4 compounds appear only in B2. In winter, these are sodium perfluoropentanesulfonate (PFPeS) and FBSA, while in spring they are 3,6-OPFHpA and PFEtS. These PFAS concentrations at B2 remain low and are linked to AFFF.[53,64,97] In contrast, downstream at B3, PFPeS and FBSA were always detected in the winter, while PFEtS was detected in the spring. However, upstream of B2, other PFAS compounds associated with AFFF such as 6:2-FTAB and PFECHS were also detected. Fecal coliform concentrations of the urban creek (U2)[98] suggest that sewer cross-connections are likely present based on indicators of sewer cross-contamination.[99] Sewer cross-connections are a potential source of PFAS compounds in urban storm sewers discharging to surface waters. FBSA and other PFAS compounds have been detected in Canadian wastewater.[88] Further study of the characteristics of urban creek U2 could help to understand its impact on Ottawa River drinking water sources as well as explain potential trends with regard to the detection of compounds typically associated with residential wastewater.

Finally, various types of chemical markers have been proposed to identify sources of fecal contamination in drinking water sources, such as the presence of caffeine, carbamazepine, or acetaminophen.[92,100] However, many of these compounds are present in low concentrations that fall below the detection limits because of degradation and dilution. Although PFAS compounds can undergo transformations, they are highly stable in the environment. As such, this study demonstrated that the signature of PFAS compounds in the St. Lawrence River was different from the signature of two other rivers in the Greater Montreal Area. PFAS compounds provide a useful tool to study large water body mixing that occurs at the confluence of rivers or the dispersion of a plume in receiving waters. Figure 4 shows clear grouping according to source waters and how mixing occurs at some of the locations. PFAS compounds can also help confirm the existence of a hydraulic connection between an urban creek draining an industrial zone with an airport and a drinking water intake. Controlled tracer studies provide the best data; however, permits to discharge substances upstream of drinking water sources are not easily obtained. While ideally there would be no PFAS contamination upstream of drinking water sources since these compounds are not removed by conventional treatment, they can provide a snapshot of contaminant flows and mixing, including at large regional scales, such as that of the Great Lakes Basin.

4. CONCLUSIONS

In response to the previously stated objectives and ensuing results and discussion, the following conclusions were drawn:

- All sampled raw water sources in the Greater Montreal Area were contaminated by PFAS substances. A total of 32 of 77 monitored PFAS compounds were detected in raw and treated drinking water for winter and spring sampling events.
- Carboxylic and sulfonic acids (PFCA and PFCA) were dominant, as expected. Several detected compounds are known precursors capable of degrading into terminal PFAS, which may contribute to temporal and spatial variations. It highlights the importance of including precursor compounds in monitoring programs and considering the transformation pathways when interpreting trends.
- Weekly sampling at a drinking water treatment plant on the St. Lawrence River from August to November 2023 demonstrated a change in the relative and total concentrations of short-chain carboxylic acids and long-chain carboxylic acids. Long-chain carboxylic acids were at their highest relative and total concentrations during and immediately following the lowest flow period of the St. Lawrence River, possibly related to basin scale dilution processes.
- Correlations among PFAS compounds revealed complex relationships influenced by compound structure, source, and environmental behavior. Legacy PFAS such as PFOS and PFOA showed strong intercorrelation due to shared sources and persistence, while more recent compounds such as 6:2-FTAB or PFECHS displayed negative correlations. Negative correlations are potentially indicative of different sources and transport pathways and degradation patterns. Contamination of drinking water sources is ongoing considering both historical and contemporary PFAS compounds, including AFFFs, wastewater discharges, and agricultural inputs.
- PFAS concentrations in raw and treated drinking water remained below recommended limits as proposed by Health Canada and the USEPA throughout the study period.
- All conventional drinking water treatment plants in the Greater Montreal Area showed limited removal of PFAS compounds, emphasizing the need for better protection of drinking water sources.
- The highest PFAS concentrations were observed in drinking water treatment plants supplied by the St. Lawrence River as compared to the Ottawa and L'Assomption Rivers despite the larger total number of wastewater and combined sewer overflow discharges upstream of the Ottawa River intakes. These results demonstrate the importance of basin scale contamination of the Great Lakes on drinking water from the St. Lawrence.
- Local scale contaminant sources were also of concern even if total concentrations did not significantly increase given the relative importance of basin scale sources of contamination. Targeted sampling of an urban creek near an airport demonstrated the role of urban runoff contributing to PFAS loads impacting downstream drinking water intakes. Regional and local scale contamination and hydrodynamic mixing in large river systems shape the temporal and spatial variability of PFAS concentrations.

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- PFAS profiles were reliable markers of the origins of the water source for intakes variably receiving both St. Lawrence and Ottawa River water and corresponded with expected results from detailed hydrodynamic studies.
- Improving PFAS mitigation strategies will require both advances in drinking water treatment and a better understanding of regional and local scale sources of contamination.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsestwater.5c00562.

Details on targeted PFAS compounds, quality control, efficiency of water treatment plants in removing PFAS compounds, the hydrographic network of Greater Montreal, and sources of PFAS contamination in Canada (PDF)

QA/QC results (XLSX)

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Notes

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